



Research Internship

Tomato Detection & Ripeness Classification using YOLOv8 Instance Segmentation

Internship report

**Submitted to D Y Patil International University, Akurdi, Pune
in partial fulfilment of full-time degree**

**B.Tech Computer Science and Engineering
(Data Science)**

Submitted By:
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20230802023

School of Computer Science Engineering and Applications

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[AY-2025-26]



This internship report is submitted for the partial fulfillment of the requirements for the award of degree of Bachelor of Technology in Computer Science and Engineering is an authentic work carried out by them under our supervision and guidance.

Internal Evaluator

Name and Sign

School of Computer Science Engineering & Applications
D Y Patil International University, Akurdi
Pune, 411044, Maharashtra, INDIA

DECLARATION

I, hereby declare that the following Internship which is being presented entitled as **Research Internship** is an authentic documentation of my own original work to the best of my knowledge. The following Internship and its report in part or whole, has not been presented or submitted by me for any purpose in any other institute or organization. Any contribution made to the internship by others, with whom I have worked at D Y Patil International University, Akurdi, Pune or elsewhere, is explicitly acknowledged in the report.

Samriddhi Pathak

20230802023

OFFER LETTER

Ref:FV/intern/2026-22

FirstVidya Pvt. Ltd

CIN NO-U80900UP2019PTC117548

Date: 03 may 2026

Company Name: FirstVidya Pvt Ltd

Company Address: C404 Star Rameshwaram, Raj Nagar Ext. Ghaziabad, UP, India



MACHINE LEARNING RESEARCH INTERN

Dear Samriddhi,

We are pleased to offer you a Research Internship at FirstVidya Pvt. Ltd. in the Research & Development (R&D) Division.

Based on your academic background and interest in AI/ML, Data Analytics, and research activities, you have been selected for this internship opportunity.

Internship Details

Position: Research Intern

Domain: AI/ML & Data Analytics

Division: Research & Development (R&D)

Mode: Remote/Hybrid

Duration: May 15, 2026 to August 5, 2026

During the internship, you will work on research-oriented tasks, data analysis, machine learning projects, and related technical activities.

Upon successful completion, you may be awarded an Internship Certificate based on your performance and participation.

We look forward to your contribution and wish you a great learning experience ahead.

Best Regards,

Priyanka K.

Director

FirstVidya Pvt. Ltd.

Priyanka K.

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Director
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Figure 0.1: Offer Letter

COMPLETION LETTER

Ref:FV/IN/2026-22

FirstVidya Pvt. Ltd
CIN NO-U80900UP2019PTC117548
Website: Firstvidya.com
Phone no: 09096210555 Email ID: firstvidya@gmail.com

Date: 5 August 2026
Company Name: FirstVidya Pvt Ltd
Company Address: C404 Star Rameshwaram, Raj Nagar Ext. Ghaziabad, UP, India

INTERNSHIP COMPLETION CERTIFICATE

Dear Samriddhi,

This is to certify that Ms. Samriddhi Pathak has successfully completed her Machine Learning Research Internship at FirstVidya Pvt. Ltd., conducted remotely from May 15, 2026 to August 5, 2026. During the 83-day internship, Samriddhi demonstrated diligence, curiosity, and strong technical acumen in the assigned tasks.

During the internship, she worked on a research-oriented machine learning project titled "Tomato Ripeness Detection and Classification using YOLOv8 Instance Segmentation." Her work involved curating and preprocessing an annotated image dataset, converting COCO-format segmentation annotations into a YOLO-compatible format, and training a YOLOv8n-seg instance-segmentation model to detect and classify tomatoes as ripe or unripe with pixel-level accuracy.

The model was rigorously evaluated using precision, recall, and mAP metrics, and further validated for generalisation on unseen real-world images. She also conducted an exploratory statistical study on the effect of background removal on downstream ripeness classification using a MobileNetV2 model.

We appreciate Samriddhi's enthusiasm, commitment, and consistent performance during the internship. We wish her continued success in all her future endeavors.

Best regards,

Priyanka K.
Priyanka K.
Director
FirstVidya Pvt. Ltd



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Figure 0.2: Completion Letter

ACKNOWLEDGEMENT

With due respect, I express my deep sense of gratitude to Research Internship Coordinators Dr. Dipika Pradhan and Dr. Somya Dubey for their valuable help and guidance. I am thankful for the encouragement that they have given in completing this Research Internship successfully.

It is imperative for me to mention the fact that the report of internship could not have been accomplished without the periodic suggestions and advice of my project supervisor (Name of supervisor).

I am also grateful to our respected Director, Prof. (Dr.) Rahul Sharma, HOD, Dr. Maheshwari S. Biradar and Hon'ble Vice Chancellor Prof. Manish Bhalla, DYPIU, Akurdi, for permitting us to carry out internship.

I am also thankful to all the other faculty members of SoCSEA ,our family members and batch mates for their kind cooperation and help.

Samriddhi Pathak (20230802023)

Abstract

Tomato ripeness assessment is an important task in precision agriculture, as it directly influences harvesting decisions, crop quality, and market value. Traditional manual inspection methods are labor-intensive, time-consuming, and prone to human error. Recent advancements in computer vision and deep learning provide opportunities to automate this process through intelligent image-based analysis.

This project presents a deep learning-based system for tomato detection and ripeness classification using YOLOv8 Instance Segmentation. The proposed approach simultaneously performs object detection, pixel-level segmentation, and classification of tomatoes into ripe and unripe categories. A COCO-annotated tomato dataset (177 images, 868 annotated instances) was used for model development, with dataset analysis, preprocessing, and visualization applied to understand data quality and characteristics.

The YOLOv8 Instance Segmentation model was trained and evaluated using Precision, Recall, and mean Average Precision (mAP) metrics, achieving a Box mAP@50 of 93.3% and a Mask mAP@50 of 93.5%. The model was further tested on previously unseen images to evaluate its generalization capability under varying environmental conditions, backgrounds, and lighting.

An additional experimental study using MobileNetV2 investigated whether removing image background via segmentation masks affects tomato ripeness classification accuracy. An initial single training run suggested a difference between classifiers trained on original versus background-removed images; however, repeating the experiment five times per condition and applying an independent samples t -test showed no statistically significant difference ($t = 0.220$, $p = 0.831$, Cohen's $d = 0.139$). This result yields both a practical and a methodological contribution. Practically, it shows that background removal can be used as a preprocessing step without a measurable accuracy trade-off. Methodologically, it demonstrates — on a concrete case — that single-run accuracy comparisons on a validation set of this size are unreliable without repetition and significance testing, an approach absent from the comparable studies reviewed in this report, which typically report single-run accuracy without verifying whether observed differences exceed normal training variance.

The proposed system demonstrates the effectiveness of deep learning and instance segmentation for automated agricultural monitoring and provides a foundation for future deployment as a real-time intelligent tomato analysis platform.

Keywords: Computer Vision, Deep Learning, YOLOv8, Instance Segmentation, Tomato Detection, Ripeness Classification, Precision Agriculture, Statistical Validation.

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1. INTRODUCTION

Agriculture plays a vital role in the global economy and food supply chain, providing food and raw materials for a growing population. Among various agricultural crops, tomatoes are one of the most widely cultivated and consumed vegetables due to their nutritional value, economic importance, and extensive use in food products. The quality and ripeness of tomatoes significantly influence harvesting decisions, market value, storage life, and consumer acceptance. Therefore, accurate assessment of tomato ripeness is an essential requirement in modern agricultural practices.

Traditionally, tomato ripeness assessment is performed manually through visual inspection by farmers and agricultural workers. While this approach is simple, it is labor-intensive, time-consuming, and highly dependent on human judgment. Factors such as varying lighting conditions, differences in fruit appearance, worker experience, and large-scale cultivation environments can lead to inconsistent evaluations and reduced efficiency. As agricultural production continues to expand, there is an increasing need for automated and intelligent systems capable of performing reliable fruit detection and ripeness assessment.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision have created new opportunities for automating agricultural monitoring tasks. Deep learning techniques, particularly Convolutional Neural Networks (CNNs), have demonstrated remarkable success in image classification, object detection, and image segmentation applications. These technologies enable machines to learn complex visual patterns directly from images, reducing the need for manual feature engineering and improving detection accuracy.

Object detection models such as YOLO (You Only Look Once) have gained significant popularity due to their ability to perform real-time object localization and classification. However, conventional object detection methods only provide bounding-box information and do not accurately capture object boundaries. In agricultural environments, tomatoes frequently overlap with one another or become partially occluded by leaves and branches, making precise object separation challenging. Instance segmentation addresses this limitation by generating pixel-level masks for individual objects, enabling more accurate localization and analysis.

This project focuses on the development of an intelligent tomato detection and ripeness classification system using YOLOv8 Instance Segmentation. The proposed approach combines object detection, segmentation, and classification within a single framework to identify tomatoes and classify them as ripe or unripe. The study includes dataset analysis, image preprocessing, model training, performance evaluation, confidence score analysis, and

generalization testing on previously unseen images. In addition, an experimental investigation is conducted to examine the effect of background removal on tomato ripeness classification using a MobileNetV2-based classifier.

The primary goal of this work is to evaluate the effectiveness of YOLOv8 Instance Segmentation for automated tomato monitoring and to explore its potential applications in precision agriculture. By providing accurate detection, segmentation, and ripeness assessment, the proposed system can contribute to improved harvesting decisions, reduced labor requirements, and enhanced agricultural productivity.

1.1. Background

Agriculture is one of the most important sectors contributing to global food production and economic development. Among various horticultural crops, tomatoes are widely cultivated and consumed due to their nutritional value and commercial importance. Accurate assessment of tomato ripeness is essential for determining the optimal harvesting time, maintaining product quality, and reducing post-harvest losses [1].

Traditionally, tomato ripeness assessment is performed manually through visual inspection by farmers and agricultural workers. However, manual inspection is labor-intensive, time-consuming, and often influenced by subjective human judgment. Variations in lighting conditions, fruit appearance, and worker experience can lead to inconsistent evaluations, particularly in large-scale agricultural operations [2].

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision have enabled the development of automated systems capable of analyzing agricultural images with high accuracy. Deep learning techniques, particularly Convolutional Neural Networks (CNNs), have demonstrated remarkable success in image classification, object detection, and segmentation tasks by automatically learning discriminative visual features from large datasets [3].

Object detection models such as Faster R-CNN, SSD, and YOLO have been widely applied to agricultural applications including fruit detection, crop monitoring, and yield estimation. Among these approaches, YOLO has gained significant popularity due to its real-time detection capability and high accuracy. Several studies have successfully utilized YOLO-based models for tomato detection under complex field conditions [4].

While object detection models provide bounding-box localization, they cannot accurately capture object boundaries. This limitation becomes significant in agricultural environments where tomatoes often overlap or are partially occluded by leaves and branches. Instance

segmentation addresses this challenge by generating pixel-level masks for individual objects, enabling more precise localization and separation of overlapping fruits [5].

The latest version of the YOLO family, YOLOv8, incorporates instance segmentation capabilities along with efficient object detection and classification. Recent studies have demonstrated that YOLOv8 Instance Segmentation can effectively detect and segment tomatoes in complex agricultural environments, making it a promising solution for automated ripeness analysis and precision agriculture applications [6].

Motivated by these developments, this project investigates the use of YOLOv8 Instance Segmentation for tomato detection and ripeness classification. The proposed system aims to provide accurate detection, pixel-level segmentation, and automated ripeness assessment while evaluating the impact of segmentation quality on downstream classification performance.

1.2. Objectives

The primary objective of this project is to develop an intelligent computer vision system capable of automatically detecting tomatoes and classifying their ripeness using deep learning techniques.

The specific objectives of the project are as follows:

1. To analyze and preprocess a COCO-annotated tomato dataset containing ripe and unripe tomato instances.
2. To visualize and examine segmentation masks, bounding boxes, and class distributions to better understand the dataset characteristics.
3. To train a YOLOv8 Instance Segmentation model for accurate tomato detection, segmentation, and ripeness classification.
4. To evaluate the performance of the trained model using standard metrics such as Precision, Recall, and Mean Average Precision (mAP).
5. To assess the generalization capability of the model on previously unseen tomato images collected from external sources.
6. To analyze prediction confidence scores and investigate the robustness of the model under varying environmental conditions.
7. To study the impact of background removal on tomato ripeness classification by conducting comparative experiments using a MobileNetV2 classifier.

8. To identify the limitations of the proposed approach and suggest possible improvements for future research and real-world deployment.
9. To establish a foundation for developing an intelligent tomato monitoring system for precision agriculture applications.

1.3. Problem statement

Tomato cultivation is an important component of modern agriculture, and accurate ripeness assessment is essential for determining the optimal harvesting time, maintaining product quality, and reducing post-harvest losses. Traditionally, tomato ripeness is evaluated through manual visual inspection, which is labor-intensive, time-consuming, and prone to human error. The effectiveness of manual assessment can be affected by factors such as lighting conditions, worker experience, and the large volume of produce that must be inspected in commercial agricultural environments.

Although computer vision and deep learning techniques have been successfully applied to agricultural monitoring, accurately detecting and classifying tomatoes remains a challenging task. Tomatoes often appear in complex field environments with varying illumination, cluttered backgrounds, overlapping fruits, and partial occlusions caused by leaves and branches. These factors can significantly reduce the performance of conventional image processing and object detection methods.

Furthermore, traditional object detection approaches provide only bounding-box localization and do not capture precise object boundaries. This limitation becomes particularly significant when multiple tomatoes overlap or appear in densely populated scenes, where accurate separation of individual fruits is required for reliable analysis.

Therefore, there is a need for an automated and intelligent system capable of accurately detecting tomatoes, generating precise segmentation masks, and classifying their ripeness under real-world agricultural conditions. The proposed project addresses this challenge by utilizing YOLOv8 Instance Segmentation to perform simultaneous tomato detection, segmentation, and ripeness classification. The system aims to improve the accuracy, reliability, and efficiency of tomato monitoring while supporting future applications in precision agriculture and automated harvesting systems.

2. LITERATURE SURVEY

2.1. Literature Review

Recent advancements in deep learning and computer vision have enabled the development of intelligent agricultural monitoring systems for fruit detection, ripeness classification, and yield estimation. Several researchers have proposed object detection and segmentation approaches for tomato recognition under complex field conditions.

Kamilaris and Prenafeta-Boldú [1] presented a comprehensive survey on the application of deep learning in agriculture. The study highlighted the effectiveness of Convolutional Neural Networks (CNNs) in agricultural image analysis, including crop classification, disease detection, and fruit recognition. The authors demonstrated that deep learning models significantly outperform traditional image processing methods by automatically learning discriminative visual features from large datasets.

Yang et al. [2] proposed a lightweight YOLOv8-based tomato detection algorithm that incorporated feature enhancement and attention mechanisms to improve detection accuracy while reducing computational complexity. Experimental results showed that the proposed model achieved high detection performance and was suitable for real-time agricultural applications. However, the method primarily focused on object detection and did not provide pixel-level segmentation information.

He et al. [5] introduced Mask R-CNN, a widely adopted instance segmentation framework capable of generating pixel-level masks in addition to object detection. The study demonstrated the advantages of instance segmentation in accurately separating overlapping objects. Although the method achieved excellent segmentation accuracy, its computational complexity limited real-time deployment in resource-constrained environments.

Wei et al. [6] proposed an instance segmentation framework for tomato ripeness detection in field environments. The model utilized segmentation masks to accurately identify tomato boundaries under varying illumination and occlusion conditions. The study reported improvements in segmentation accuracy compared to conventional object detection approaches. However, the influence of segmentation quality on downstream classification tasks was not investigated.

Song et al. [7] developed FGS-YOLOv8s-Seg, a lightweight instance segmentation model for tomato maturity detection in greenhouse environments. Their work demonstrated that instance segmentation could improve maturity classification performance while maintaining

computational efficiency. Despite these improvements, challenges remained in densely populated scenes containing overlapping tomatoes and partial occlusions.

Although existing studies have achieved promising results in tomato detection and segmentation, most research primarily focuses on improving detection accuracy and computational efficiency. Limited attention has been given to understanding how segmentation quality influences downstream classification tasks and whether contextual information contributes to ripeness prediction. Furthermore, many studies evaluate performance only on benchmark datasets without extensive analysis of model generalization on previously unseen images. These limitations motivated the present work, which combines YOLOv8 Instance Segmentation with additional experiments on background removal, classification performance, and real-world generalization.

Overall, the reviewed studies demonstrate that deep learning and instance segmentation have significantly improved tomato detection and maturity analysis. However, several limitations continue to exist in current approaches.

2.2. Drawbacks of existing system

The literature survey revealed several limitations in existing tomato detection and ripeness classification systems:

- Many existing approaches rely solely on object detection and provide only bounding-box localization without accurate object boundaries.
- Detection accuracy often decreases in the presence of overlapping fruits, dense foliage, and partial occlusion.
- Several studies focus primarily on detection performance and do not investigate the impact of segmentation quality on downstream classification tasks.
- Existing methods are commonly evaluated on controlled datasets and may exhibit reduced performance under varying environmental conditions.
- Limited research has been conducted on understanding how background information influences tomato ripeness classification performance.

2.3. Gaps Identified

1. Gap 1: Lack of Pixel-Level Analysis in Traditional Detection Systems

Many tomato detection systems rely on bounding-box detection and do not provide precise object boundaries. This limits their effectiveness in crowded agricultural environments where accurate separation of overlapping tomatoes is required.

2. Gap 2: Limited Investigation of Segmentation Quality

Although instance segmentation methods improve object localization, few studies analyze how segmentation quality affects downstream classification performance and model reliability.

3. Gap 3: Insufficient Generalization Analysis

Several existing studies evaluate models only on validation datasets and provide limited evidence regarding their performance on previously unseen real-world images.

4. Gap 4: Limited Study of Contextual Information

Current research rarely investigates whether removing background information improves or degrades tomato ripeness classification performance.

2.4. Objective

Based on the identified research gaps, this project aims to develop a YOLOv8 Instance Segmentation-based tomato detection and ripeness classification system capable of generating pixel-level object masks, achieving robust performance under real-world conditions, and investigating the influence of segmentation quality and background removal on downstream classification tasks. In addition to the four gaps identified above, this project also addresses a methodological gap common across the reviewed studies: none of the surveyed works ([2], [5], [6], [7]) report repeated-run statistical validation when comparing model or preprocessing variants, relying instead on single-run accuracy figures.

3. PROPOSED METHODOLOGY

3.1. PROPOSED METHODOLOGY

The proposed system aims to automatically detect tomatoes and classify their ripeness using YOLOv8 Instance Segmentation. The methodology consists of multiple stages including dataset analysis, image preprocessing, model training, performance evaluation, and experimental analysis. The overall workflow begins with a COCO-annotated tomato dataset containing images, segmentation masks, and ripeness labels.

Initially, the dataset was analyzed to understand class distribution, object frequency, bounding box dimensions, and segmentation mask characteristics. The dataset was then split into training, validation, and test sets. During training, the YOLOv8 framework's built-in augmentation pipeline (including mosaic augmentation, horizontal flipping, and HSV-based color jittering) was used with its default settings to improve model robustness and generalization capability; no additional, separately implemented augmentation stage was applied.

The processed dataset was used to train a YOLOv8 Instance Segmentation model. The model simultaneously performed object detection, segmentation, and ripeness classification by identifying tomatoes and classifying them as ripe or unripe. During training, model performance was monitored using Precision, Recall, and Mean Average Precision (mAP) metrics.

After training, the model was evaluated on validation data and tested on previously unseen images to assess its generalization capability. Confidence score analysis was also conducted to examine prediction reliability.

Furthermore, an experimental study was performed using MobileNetV2 to investigate the effect of background removal on ripeness classification. To ensure the resulting comparison was statistically reliable rather than an artifact of a single training run, each classifier (trained on original images and on background-removed images) was retrained five times independently, and the results were compared using a two-sample t -test and effect size (Cohen's d).

The overall methodology adopted in this work is illustrated in Figure 3.1.

3.2. Block diagram

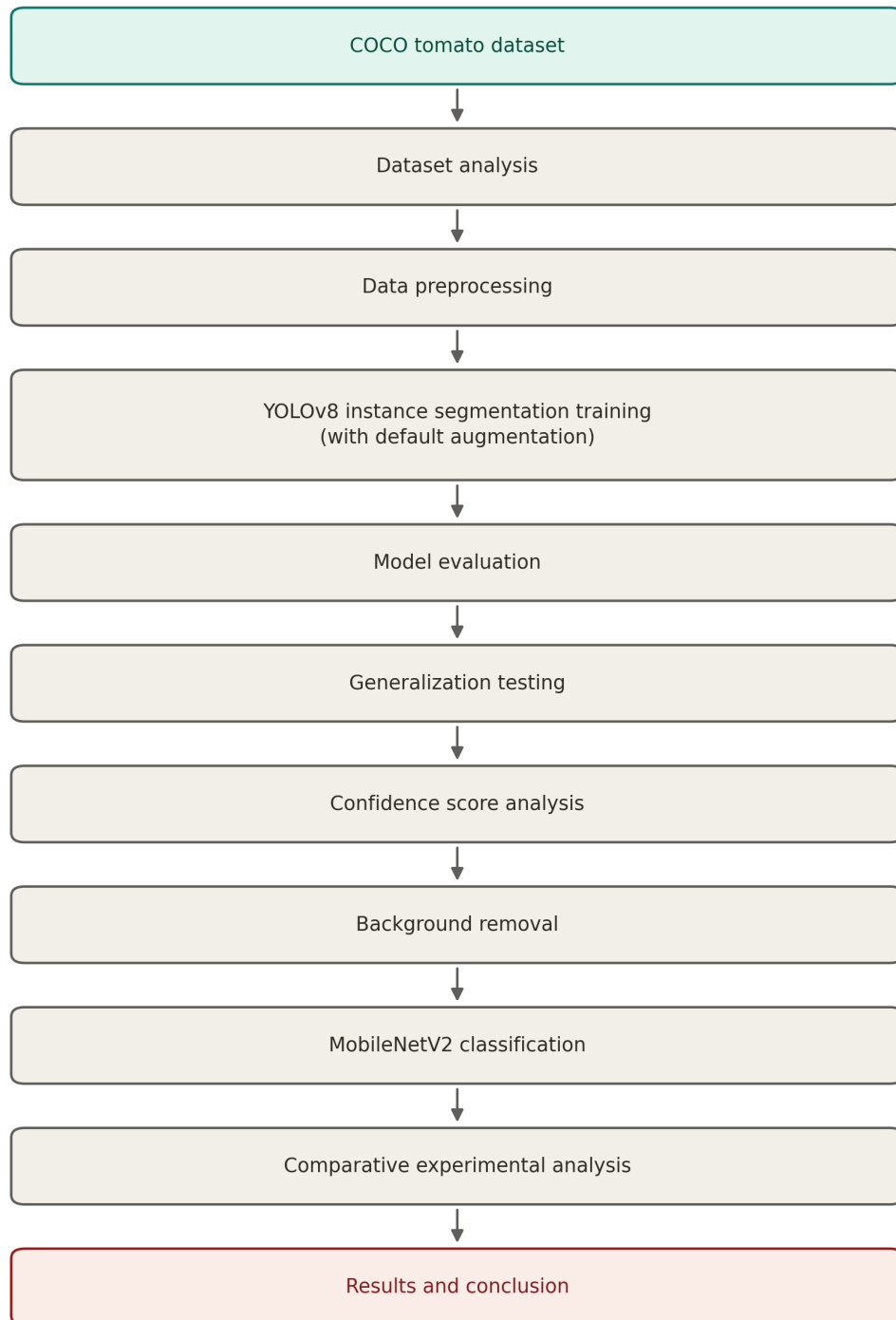


Figure 3.1: Overall methodology of the proposed tomato detection and ripeness classification system

3.3. Tools Used

The following software tools, libraries, and frameworks were utilized during the development of the proposed system:

- Python – Primary programming language used for implementation.
- Google Colab – Cloud-based environment used for model development and training.
- YOLOv8 (Ultralytics) – Deep learning framework used for tomato detection and instance segmentation.
- TensorFlow/Keras – Used for implementing and training the MobileNetV2 classification model.
- OpenCV – Used for image processing and visualization tasks.
- NumPy – Used for numerical computations and array operations.
- Pandas – Used for dataset analysis and tabular data processing.
- Matplotlib – Used for data visualization and performance analysis.
- COCO Dataset Format – Used for storing image annotations, segmentation masks, and class labels.

4. ANALYSIS AND DESIGN

4.1. Dataset Analysis

The dataset used in this study consists of tomato images annotated in COCO format for instance segmentation. Each image contains segmentation masks, bounding boxes, and class labels corresponding to ripe and unripe tomatoes. Dataset analysis was performed to understand class distribution, object frequency, and annotation quality before model training.

The analysis revealed a balanced distribution of ripe and unripe tomato instances, making the dataset suitable for supervised learning. Additionally, visualization of segmentation masks and bounding boxes confirmed the correctness of annotations and helped identify challenging samples containing overlapping tomatoes and occlusions.

4.2. Dataset Statistics

The overall dataset statistics are presented in Table 5.1.

4.3. System Architecture

The overall architecture of the proposed tomato detection and ripeness classification system is illustrated in Figure 4.1. The workflow begins with dataset acquisition and preprocessing, followed by YOLOv8 Instance Segmentation training. The trained model is evaluated using standard performance metrics and further analyzed through additional experiments.

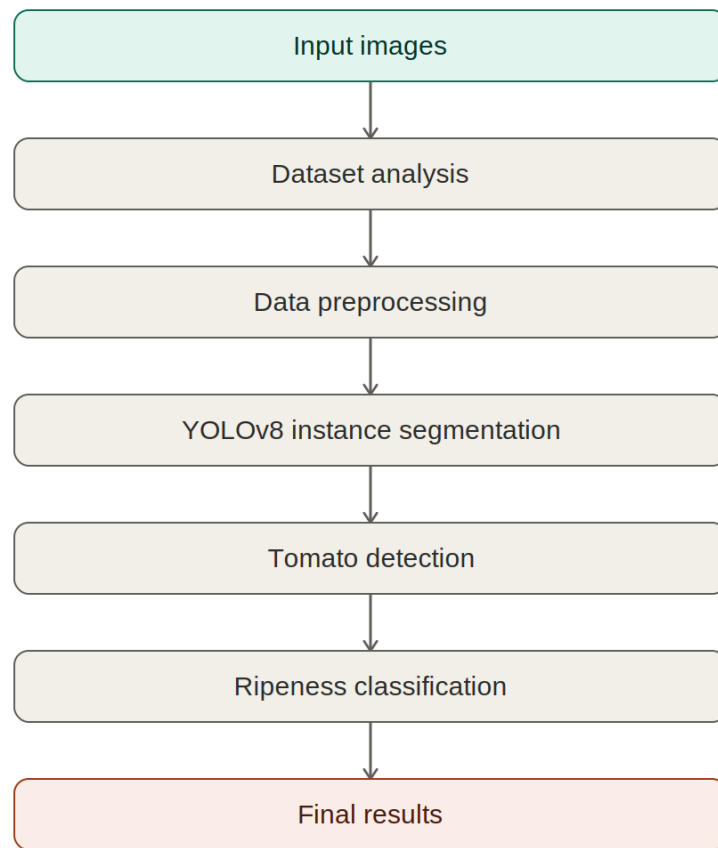


Figure 4.1: Overall system architecture of the proposed approach

4.4. Proposed Workflow

The proposed workflow consists of dataset analysis, image preprocessing, model training, evaluation, and experimental analysis. Initially, the dataset is analyzed to understand its characteristics. Standard YOLOv8 training augmentation was used. The processed data is used to train a YOLOv8 Instance Segmentation model for tomato detection and ripeness classification. Finally, the trained model is evaluated and compared using multiple experimental studies.

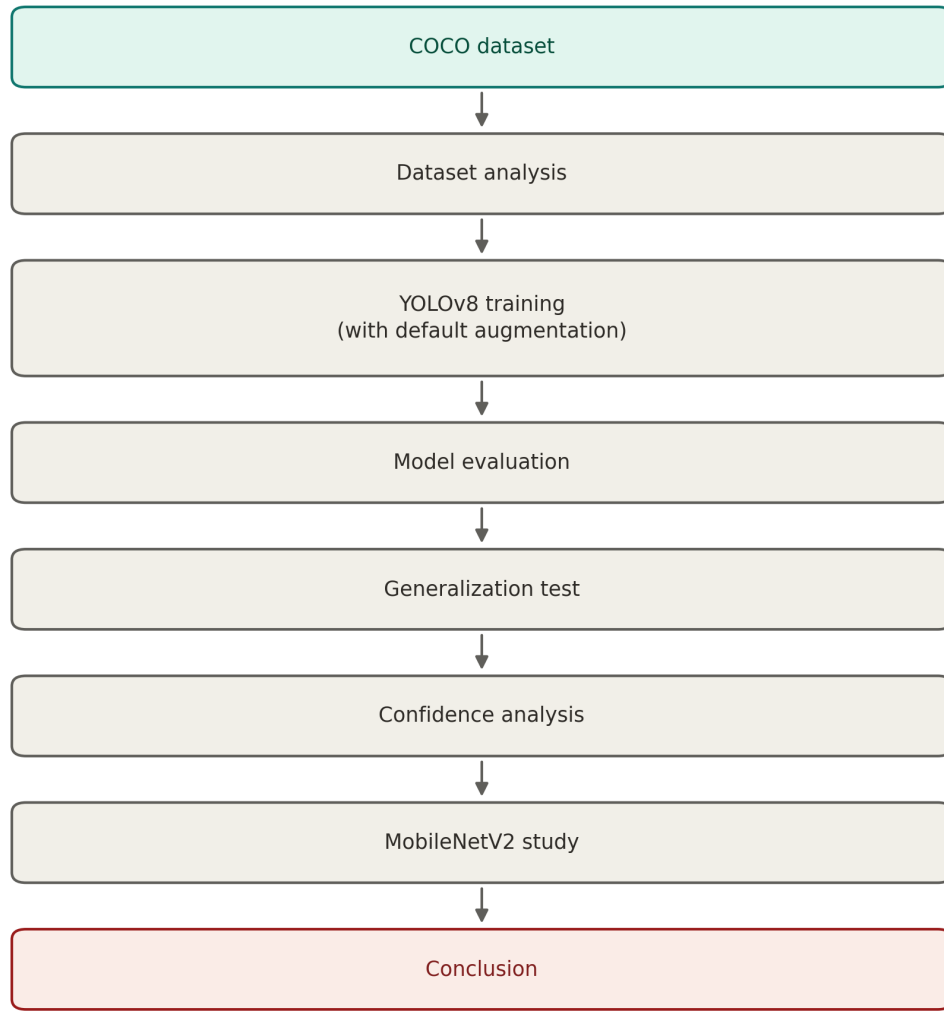


Figure 4.2: Workflow of the proposed tomato detection and ripeness classification system

4.5. YOLOv8 Instance Segmentation Architecture

YOLOv8 Instance Segmentation was selected due to its ability to perform object detection, segmentation, and classification simultaneously. The architecture consists of a backbone network for feature extraction, a neck for feature aggregation, and a segmentation head that generates pixel-level masks for each detected object.

Figure 4.3 illustrates the architecture of the YOLOv8 Instance Segmentation model used in this study.

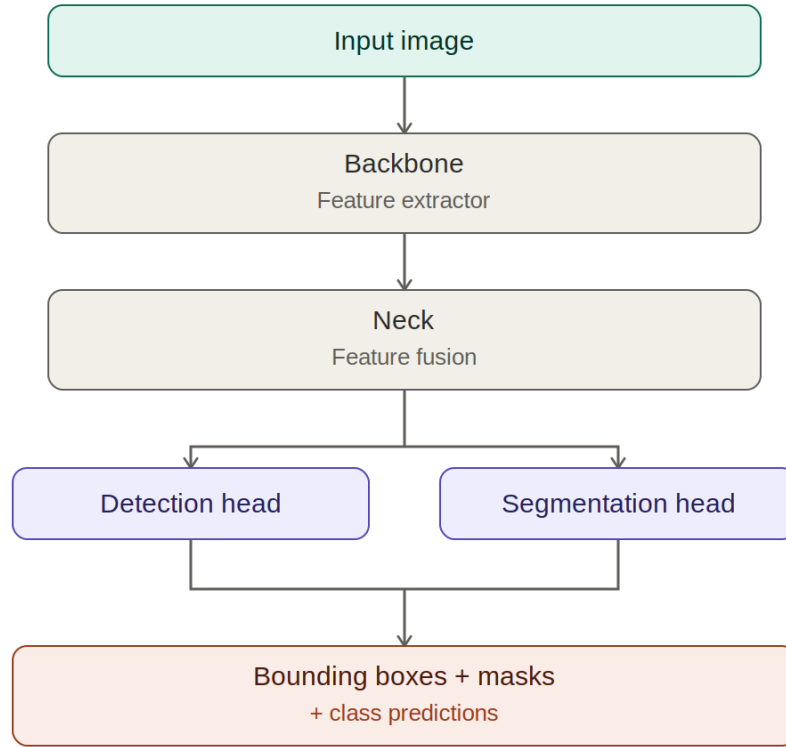


Figure 4.3: YOLOv8 Instance Segmentation Architecture

4.6. Experimental Design

To evaluate the effectiveness of the proposed approach, the following experiments were conducted:

1. Training and evaluation of the YOLOv8 Instance Segmentation model using the annotated tomato dataset.
2. Generalization testing using previously unseen tomato images.
3. Confidence score analysis to evaluate prediction reliability.
4. Comparative classification study using MobileNetV2 on original tomato images.
5. Comparative classification study using MobileNetV2 on background-removed tomato images generated from segmentation masks.
6. A robustness check repeating experiments 4 and 5 across five independent training runs each, followed by an independent samples t -test and effect-size (Cohen's d) calculation,

to determine whether any observed difference between conditions is statistically meaningful or attributable to normal training variance.

7. Diagnostic analysis of segmentation mask quality and its relationship to classification correctness, to check for a confound that a simple accuracy comparison might miss.

These experiments provide a comprehensive evaluation of both the segmentation model and the downstream classification performance.

4.7. Advantages & Disadvantages

Advantages

- Performs simultaneous tomato detection, segmentation, and ripeness classification within a single framework.
- Generates pixel-level segmentation masks for accurate object localization.
- Handles overlapping tomatoes more effectively than traditional bounding-box-only object detection.
- Demonstrates good generalization capability on previously unseen images.
- Background removal can be applied as a preprocessing step (e.g. for size estimation or further isolation of the fruit) without a measurable classification accuracy trade-off, as confirmed by repeated-run statistical testing (Section 5.7).
- Can be extended for real-time agricultural monitoring and harvesting applications.

Disadvantages

- Model performance depends heavily on annotation and segmentation mask quality.
- Detection confidence may decrease under severe occlusion and dense foliage.
- Training requires significant computational resources and GPU acceleration.
- Evaluated on a comparatively small dataset (177 images, 868 instances); conclusions regarding generalization should be validated on larger datasets before deployment.
- Performance can be affected by extreme lighting variations and highly cluttered environments.

5. RESULTS AND DISCUSSIONS

5.1. Dataset Analysis Results

The dataset analysis revealed that the tomato dataset contained a total of 177 images and 868 annotated tomato instances. Among these instances, 429 were labeled as ripe and 439 were labeled as unripe, indicating a balanced class distribution suitable for supervised learning. The average number of tomatoes per image was approximately 4.90.

The balanced distribution of classes reduced the risk of model bias toward a particular category and provided a strong foundation for training the YOLOv8 Instance Segmentation model.

Table 5.1: Tomato Dataset Statistics

Parameter	Value
Total Images	177
Total Tomatoes	868
Ripe Tomatoes	429
Unripe Tomatoes	439
Average Tomatoes per Image	4.90

The overall dataset distribution is presented in Table 5.1.

5.2. YOLOv8 Training Results

The YOLOv8 Instance Segmentation model was trained using the annotated tomato dataset. During training, the model progressively improved its ability to detect, segment, and classify tomatoes. Training and validation loss curves indicated stable convergence without significant overfitting.

The trained model achieved high detection and segmentation performance as measured by Precision, Recall, and Mean Average Precision (mAP). These results demonstrate the effectiveness of YOLOv8 Instance Segmentation for tomato detection and ripeness classification tasks.

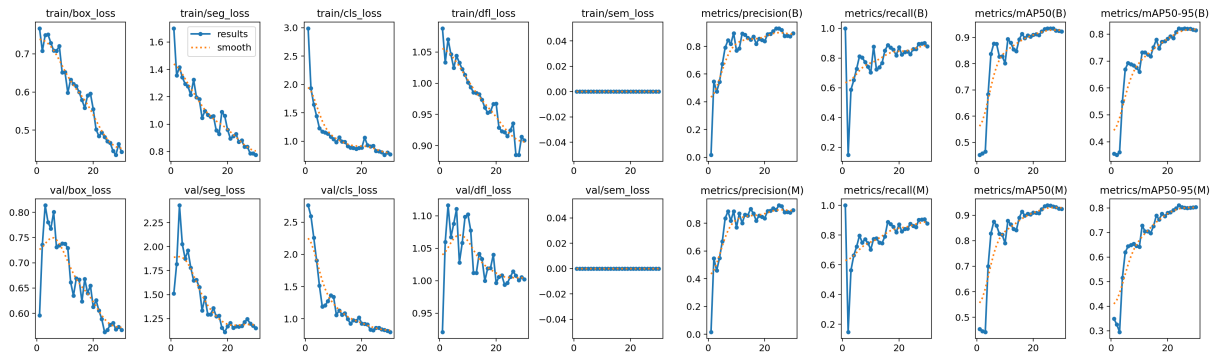


Figure 5.1: YOLOv8 training and validation performance

5.3. Model Evaluation Results

Model evaluation was conducted using standard object detection and segmentation metrics including Precision, Recall, and Mean Average Precision (mAP). The proposed YOLOv8 Instance Segmentation model achieved a Box mAP@50 of 93.3%, a Mask mAP@50 of 93.5%, a Precision of 93.0%, and a Recall of 82.5%.

The obtained results demonstrate strong detection and segmentation performance for both ripe and unripe tomatoes. The high precision value indicates that the majority of predicted detections were correct, while the strong mAP scores confirm accurate localization and segmentation of tomato instances. Furthermore, the evaluation metrics indicate that the model successfully learned discriminative features from the training dataset and was capable of reliably detecting and classifying tomatoes under varying conditions.

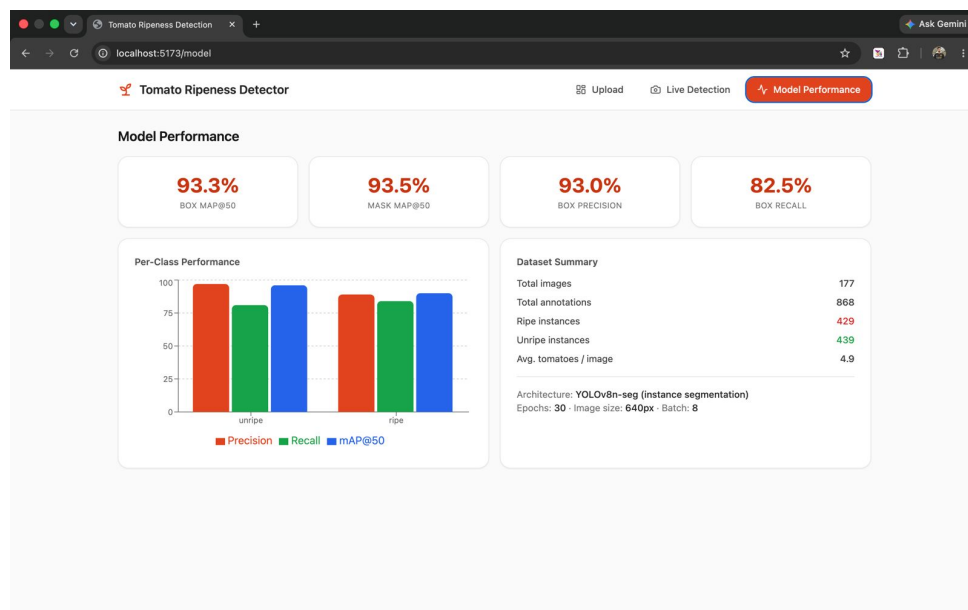


Figure 5.2: Performance evaluation of the proposed YOLOv8 Instance Segmentation model

5.4. Generalization Experiment

To evaluate the model’s ability to generalize beyond the training dataset, four tomato images collected from Google Images were used for inference. These images were completely unseen during training and contained different backgrounds, lighting conditions, viewpoints, and levels of tomato overlap.

The prediction results demonstrated that the trained YOLOv8 Instance Segmentation model successfully detected and classified tomatoes in all test images. The model identified both ripe and unripe tomatoes under challenging environmental conditions while maintaining reliable detection performance.

These findings indicate that the proposed model possesses good generalization capability and can be applied to previously unseen tomato images with satisfactory performance.

Table 5.2: Generalization Test Results

Image	Total Detected	Ripe	Unripe	Observation
image1.jpg	5	4	1	Successful detection
image2.jpg	30	19	11	Successful detection
image4.jpg	12	9	3	Successful detection
image5.jpg	5	5	0	Successful detection

Table 5.2 shows the detection results obtained on previously unseen tomato images. The model successfully identified tomato instances under varying environmental conditions, demonstrating good robustness and generalization capability.

5.5. Confidence Score Analysis

Confidence score analysis was performed using the prediction outputs generated by the YOLOv8 model. The confidence values indicate the level of certainty associated with each detection.

The majority of detections were associated with high confidence values, demonstrating the reliability of the trained model. Only a small number of detections exhibited lower confidence scores, primarily in cases involving occlusion, overlapping tomatoes, or complex backgrounds.

Overall, the confidence score distribution suggests that the model produces robust and trustworthy predictions across different image conditions.

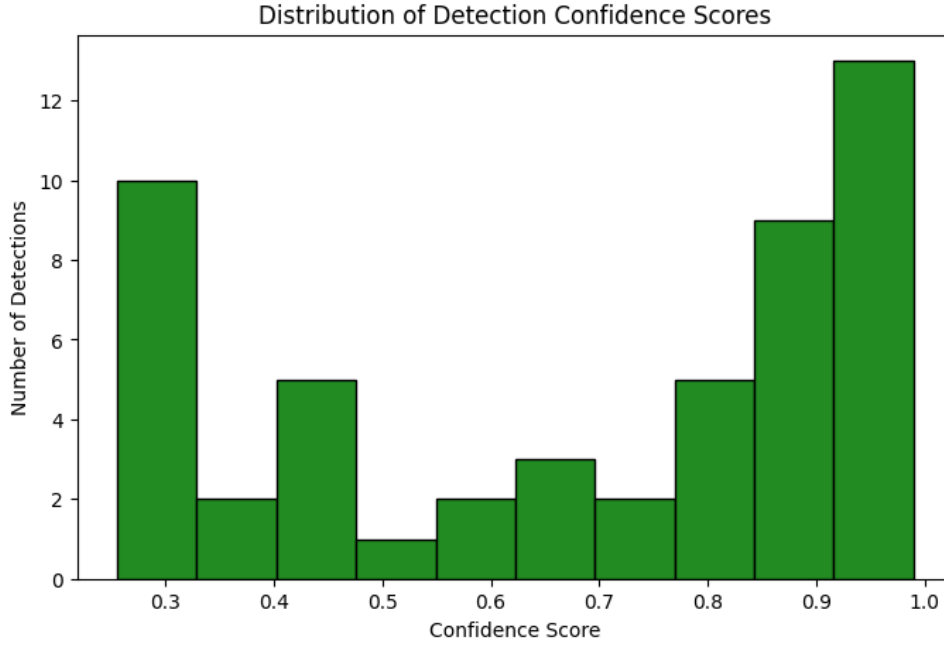


Figure 5.3: Distribution of prediction confidence scores

Figure 5.3 illustrates the distribution of confidence scores generated by the YOLOv8 model. Most detections achieved high confidence values, indicating reliable prediction performance.

5.6. Background Removal Experiment

An additional experiment was conducted to investigate whether background removal could improve tomato ripeness classification performance. Segmentation masks generated by the YOLOv8 model were used to create background-removed tomato images. A MobileNetV2 classifier was then trained on both the original images and the background-removed images.

An initial single training run suggested a difference between the two conditions. However, given the modest size of the validation set, a single run is not sufficient to determine whether such a difference reflects a real effect of background removal or ordinary training variance. To test this, each classifier was retrained independently five times, using a different random initialization and train/validation split for each run, and validation accuracy was recorded for every run.

Table 5.3 summarises the results across the five runs for each condition.

The mean accuracy of the two conditions differs by less than one percentage point, which is smaller than either condition’s own run-to-run standard deviation, indicating the apparent difference seen in a single run is within normal training variance.

Table 5.3: Classification Accuracy Across Repeated Training Runs (n = 5 per condition)

Dataset	Mean Accuracy (%)	Std. Dev. (%)	Min (%)	Max (%)
Original Images	85.79	2.78	82.71	89.25
Background Removed Images	85.33	3.85	80.37	90.19

5.7. Comparative Analysis

To formally determine whether the observed difference in mean accuracy between the two conditions is statistically meaningful, an independent samples t -test was conducted on the five accuracy values obtained for each condition. The effect size was additionally quantified using Cohen’s d . The results are summarised in Table 5.4.

Table 5.4: Independent Samples t -test: Original vs. Background-Removed Images

Statistic	Value
t -statistic	0.220
p -value	0.831
Cohen’s d	0.139

The t -test yields $p = 0.831$, which is far above the conventional significance threshold of $\alpha = 0.05$. This indicates that the null hypothesis of no difference between the two conditions cannot be rejected. The effect size, Cohen’s $d = 0.139$, is also negligible by standard conventions (below the threshold of 0.2 typically used for even a “small” effect), confirming that any residual difference between conditions is unlikely to be practically meaningful.

Beyond the specific finding that background removal does not affect classification accuracy, this experiment illustrates a broader contribution of this work: demonstrating, on a concrete case, how a single-run comparison can produce a misleading conclusion. The initial single run suggested a meaningful gap between conditions; only by repeating each condition five times and applying a two-sample t -test and Cohen’s d was it possible to show that the gap fell within normal training variance. Follow-up diagnostic checks (mask-quality analysis, failure-case inspection) further ruled out a hidden confound rather than assuming the null result was final. This repeated-run, diagnostics-backed validation protocol is not standard practice in the tomato detection/classification literature reviewed in Chapter 2, and is offered here as a reusable methodology for evaluating preprocessing choices on small validation sets — a common constraint in agricultural computer vision, where annotated datasets are often limited in size (as in this study, 177 images).

Taken together, these results indicate that background removal does *not* produce a statistically significant or practically meaningful change in tomato ripeness classification accuracy for this dataset. This finding differs from the conclusion suggested by a single training run, and highlights the importance of repeated trials and significance testing when evaluating

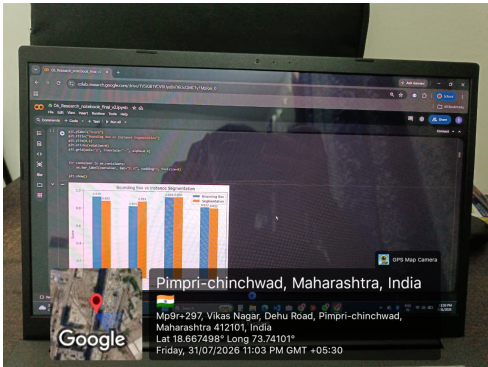
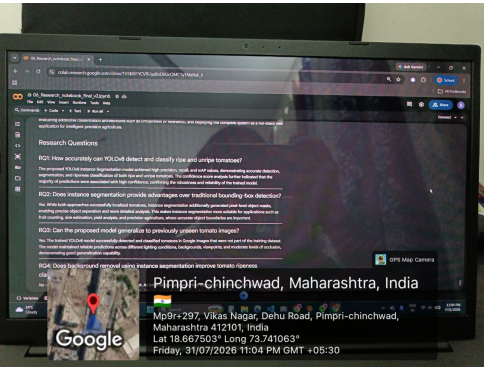
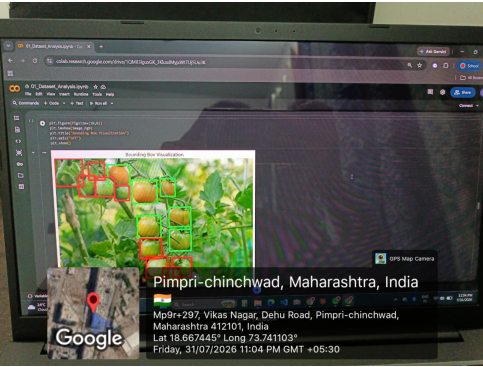
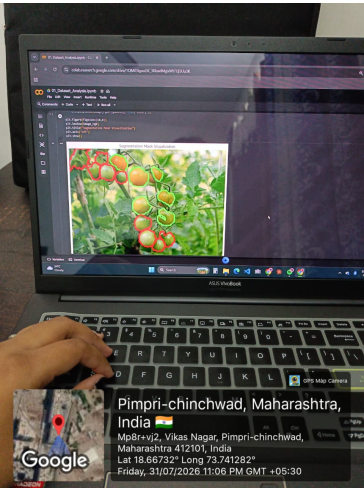
preprocessing choices on a validation set of limited size.

5.8. Limitations

Several limitations should be considered when interpreting the results of this study.

- **Dataset size.** The dataset used for both the segmentation and classification experiments (177 images, 868 annotated instances) is comparatively small. While results were consistent across repeated runs, larger-scale validation would strengthen confidence in the model’s generalization.
- **No cross-validation.** A single train/validation/test split was used for the YOLOv8 segmentation model. k -fold cross-validation was not performed due to time and compute constraints, and would provide a more robust estimate of segmentation performance.
- **Classification validation set size.** The MobileNetV2 classification experiments used a validation split of approximately 214 images. As demonstrated in Section 5, accuracy on a set of this size can vary by several percentage points between otherwise identical training runs, which is why repeated runs and significance testing were necessary before drawing conclusions.
- **Segmentation mask quality.** Some annotated instances had incomplete or imperfect segmentation masks, which could affect both the background-removal experiment and downstream classification, though the diagnostic analysis in this study did not find this to be a significant factor.

6. GEO TAGGED ON FIELD/REMOTE PHOTOS



7. CONCLUSION

This project presented an intelligent tomato detection and ripeness classification system based on YOLOv8 Instance Segmentation. The proposed approach successfully combined object detection, instance segmentation, and ripeness classification within a unified deep learning framework. Through dataset analysis, preprocessing, model training, evaluation, and experimental analysis, the effectiveness of the proposed system for precision agriculture applications was thoroughly investigated.

The experimental results demonstrated that the YOLOv8 Instance Segmentation model achieved strong detection and segmentation performance, accurately identifying both ripe and unripe tomatoes in complex agricultural environments. The model exhibited high precision, recall, and mean Average Precision (mAP) values, indicating reliable localization and classification capabilities. Furthermore, evaluation on previously unseen images confirmed the model’s ability to generalize beyond the training dataset and maintain satisfactory performance under varying lighting conditions, backgrounds, viewpoints, and levels of occlusion.

An additional experimental study was conducted to investigate the impact of background removal on tomato ripeness classification using a MobileNetV2 classifier. An initial single-run comparison suggested a difference between classifiers trained on original and background-removed images. However, repeating each experiment five times and applying an independent samples t -test showed that this apparent difference (mean accuracy 85.79% for original images versus 85.33% for background-removed images) was **not statistically significant** ($t = 0.220$, $p = 0.831$), with a negligible effect size (Cohen’s $d = 0.139$). Follow-up diagnostic analyses – including failure case visualization and mask quality analysis – found no evidence that segmentation mask quality was associated with classification errors, reinforcing the conclusion that background removal has no meaningful effect on downstream classification accuracy for this dataset.

This is itself a useful and practically relevant finding, and the validation methodology behind it constitutes the primary methodological contribution of this project: a repeated-run, statistically tested, diagnostically verified protocol for evaluating preprocessing choices under small-dataset conditions — an approach absent from the reviewed literature, which typically reports single-run comparisons without significance testing. instance segmentation-based background removal can be used as a preprocessing step, for example for size estimation or further isolation of the fruit, without incurring a measurable classification accuracy trade-off. It also illustrates a broader methodological point – single-run accuracy comparisons on a validation set of limited size can be misleading, and conclusions of this kind should be corroborated with repeated trials and formal significance testing before being reported. Section 5.8 discusses the dataset-size

and validation-set-size constraints that make this methodological caution particularly relevant to this study.

The confidence score analysis further demonstrated that the majority of predictions were associated with high confidence values, confirming the robustness and reliability of the trained segmentation model.

Overall, the results demonstrate that YOLOv8 Instance Segmentation is an effective solution for automated tomato detection and ripeness analysis. The proposed system provides accurate object localization, pixel-level segmentation, and reliable ripeness classification, making it suitable for applications in precision agriculture, crop monitoring, yield estimation, and intelligent harvesting systems. Future work may focus on validating the background-removal finding on a larger dataset, improving segmentation quality through additional annotation refinement, evaluating additional classification architectures such as EfficientNet or ResNet50, and developing a real-time web-based agricultural monitoring system.

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